

B.4 STOCHASTIC WEIGHT PRUNING (SWP)

Observing the failure of deterministic weight pruning, we tested a mix of stochasticity and pruning, the stochastic weight pruning method. Very similar to the idea of SAP, we consider all the entries of a weight tensor to be a multinomial distribution, and we sample from it with replacement. For a weight tensor $W^i \in \mathbb{R}^{a^i}$, we sample from it r^i times with replacement. The probability of sampling $(W^i)_j$ is given by

$$p_j^i = \frac{|(W^i)_j|}{\sum_{k=1}^{a^i} |(W^i)_k|}.$$

The new weight entry, $M(W^i)_j$, is given by

$$M(W^i)_j = W_j^i \cdot \frac{\mathbb{I}(W_j^i)}{1 - (1 - p_i)^{r^i}},$$

where $\mathbb{I}(W_j^i)$ is the indicator function that returns 1 if W_j^i was sampled at least once, and 0 otherwise.

For these experiments, for each weight matrix $W^i \in \mathbb{R}^{m^i}$, the number of samples picked were $k\%$ of m^i . Since samples were picked with replacement, k could be more than 100. We will refer to k as the percentage of samples drawn.

These models behave very similar to the dense model. We tried drawing range of percentages of samples, but no evident robustness could be seen against adversarial examples (Figure 5d, the legend indicates the value of k). For a small s , it is very similar to the dense model. As s increases, the these models do marginally better for low non-zero λ values, and then drops again (similar to the SAP case).

B.5 RANDOM NOISY ACTIVATIONS (RNA)

Next we change our attention to the activation maps in the dense model. One simple way of introducing stochasticity to the activations is by adding random Gaussian noise to each activation entry, with mean 0 and constant standard deviation, s . So each activation map h^i now changes to $M(h^i)$, where the j 'th entry is given by

$$M(h^i)_j = (h^i)_j + \eta, \quad \eta \sim \mathcal{N}(0, s^2).$$

These models too do not offer any robustness against adversarial examples. Their accuracy drops quickly with λ and s (Fig. 5e, the legend indicates the value of s).

B.6 RANDOMLY SCALED ACTIVATIONS (RSA)

Instead of having additive noise, we can also make the model stochastic by scaling the activations. The scale factor can be picked from a Gaussian distribution, with mean 1 and constant standard deviation s . So each activation map h^i now changes to $M(h^i)$, where the j 'th entry is given by

$$M(h^i)_j = \eta(h^i)_j, \quad \eta \sim \mathcal{N}(1, s^2).$$

These models perform similar to the dense model, exhibiting no additional robustness against adversarial examples (Figure 5f, the legend indicates the value of s).